Mughesh Kumar N R

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Education

University of Virginia *Aug 2025 – Dec 2026*
Master of Science in Computer Science Charlottesville, VA
- **Coursework:** Generative AI & LLMs, Machine Learning, Data Engineering Pipelines, Statistical Modeling.
- Graduate Teaching Assistant: Mentored and instructed 50+ students in hands-on data science labs, breaking down complex mathematical algorithms, Python/SQL workflows, and model evaluation metrics into accessible concepts.
- Judged collegiate hackathons and co-led campus AI research initiatives promoting ethical AI governance and context groundedness.

University of California, Berkeley *Jan 2025 – May 2025*
Startup Semester Program Berkeley, CA
- Completed intensive immersion in Product Management, Data-Driven Go-to-Market Strategy, and AI Business Commercialization, turning analytical findings into strategic, executive-level business decisions.
- Awarded **$10,950** UC Berkeley Chancellor's Scholarship for Startup Semester, selected from 600+ international applicants.

SRM Institute of Science and Technology *Aug 2021 – Dec 2024*
Bachelor of Technology in Computer Science and Engineering (AI & ML) Chennai, India
- Awarded University Gold Medal for Academic Excellence; elected Student Body President representing 20,000+ peers, partnering with university executives on institutional data initiatives.
- Won two international SDG hackathons and **$1,500** in prizes, developing predictive data solutions for industry juries.
- Published two peer-reviewed IEEE research papers and granted an Indian patent for data-driven IoT telemetry systems.

Work Experience

University of Virginia Information Technology Services (ITS) *June 2026 – Aug 2026*
Data Analytics & Systems Deployment Intern Charlottesville, VA
- Partnered with enterprise IT engineers on a **$6M+ Campus Infrastructure** project, analyzing telemetry from 6,656 network devices to optimize performance, resource allocation, and operational reliability.
- Architected automated **Python (Pandas, NumPy)** and **SQL** data pipelines to extract, clean, and structure unstructured system logs, reducing data preparation latency and query turnaround by 35%.
- Engineered geospatial telemetry dashboards in ArcGIS and Python, translating complex operational metrics and network health into actionable, documented data products for university leadership.
- Implemented automated data validation checks and anomaly detection routines to continuously monitor data quality, ensuring high groundedness and reliability for downstream operational analytics.

IBM *June 2023 – July 2023*
Data Science & AI Intern India
- Engineered automated end-to-end data pipelines in **Python** and **SQL** across 100,000+ records of structured and unstructured health data, modeling predictive life-years metrics with automated validation checks.
- Developed and evaluated predictive machine learning models in Scikit-learn, achieving 99.4% R^2 accuracy and validating model quality metrics (accuracy, latency, groundedness) against baseline benchmarks.
- Synthesized statistical models and feature insights into documented, supportable data products, interactive Tableau dashboards, and technical documentation, communicating trade-offs to business partners.

Projects

FaceMark AI — Biometric Feature Extraction & Vector Analytics [Live Demo]
Tech Stack: *Python, JavaScript, WebRTC, MTCNN, Vector Embeddings, REST APIs, Vercel.*
- Architected a browser-based biometric analytics platform generating 128-D vector embeddings for high-dimensional feature representation, facial verification, and real-time identity matching.
- Designed metadata filtering and retrieval routines, optimizing cosine similarity search to achieve sub-second inference latency, high accuracy, and 98.7% verification precision across varied lighting.
- Implemented automated data quality checks, voice prompts, and CSV analytical export pipelines, delivering an auditable, documented, and supportable edge data product without cloud dependencies.

Agri-Vision: Edge AI & Predictive Data Pipeline for Crop Diagnostics
Tech Stack: *Python, TensorFlow, OpenCV, SQL, Pandas, IoT Telemetry, NVIDIA Jetson Nano.*
- Developed an end-to-end computer vision and predictive data pipeline on 12,000+ field-annotated images, classifying 8+ agricultural anomalies with 96% diagnostic accuracy for regional farm operations.
- Processed unstructured image data and structured environmental sensor streams using Python and SQL, engineering contextual metadata features and automated data hygiene checks for model inference.
- Evaluated model performance across latency, compute cost, and accuracy, documenting technical trade-offs, diagnostic evaluations, and context engineering best practices for external agricultural partners.

Technical Skills

Data Science & ML: GenAI, LLMs, Retrieval-Augmented Generation (RAG), Vector Embeddings, Statistical Modeling.
Languages & Tools: Python (Pandas, NumPy, Scikit-learn, PyTorch), SQL (PostgreSQL, MySQL), C/C++, R, Bash, Git.
Evaluations & Context: Context Engineering, Prompt Tuning, Retrieval Quality, Groundedness, Latency, CI/CD, Agile.
Data Platforms & Cloud: Tableau, Power BI, ArcGIS, REST APIs, Docker, Linux, AWS (S3, EC2), Google Cloud Platform.

Publications & Patents

- Energy Efficiency Optimization in Smart Street Lighting Systems via LoRaWAN IoT – IEEE [Publication]
- Handwritten Digit Recognition Comparative Analysis with CNN & LeNet-5 Models – IEEE [Publication]
- IoT-Based Attendance and Monitoring System for Remote Operations Sites – IP India Granted [Patent]